

THE NULL DISTRIBUTIONS OF SOME TEST CRITERIA OF MULTIVARIATE NORMALITY

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SUMMARY

For testing the hypothesis that the sample of p -dimensional vectors $X_1, \dots, X_n$ come from a multivariate normal distribution the null distributions of test statistics of the form $\sup T(a'X_1, \dots, a'X_n)$ (the sup being over p -vectors $a \neq 0$) are studied where $T(V_1, \dots, V_n)$ is a test statistic for univariate normality; the choices of T considered are third moment skewness, fourth moment kurtosis and a quadratic measure of distance between the order statistics and their expected values.

1. Introduction

Let $X_1, \dots, X_n$ be i.i.d. p -dimensional (column) random vectors having common distribution function F and suppose the hypothesis $H_0: F = N_p(\mu, \Sigma)$ for some unspecified μ and Σ is to be tested. Here $N_p(\mu, \Sigma)$ denotes the p -variate normal distribution with mean μ and covariance matrix Σ .

One possible approach to obtaining test criteria for this purpose is this: let $T(V_1, \dots, V_n)$ be a test statistic for the 1-dimensional case with observations $V_1, \dots, V_n$ and suppose T is chosen so that large values are considered significant; then since X_i will be $N_p(\mu, \Sigma)$ distributed if and only if $a'X_i$ is $N_1(a'\mu, a'\Sigma a)$ for all non-zero p -dimensional real vectors a (a' denoting transpose), a plausible test statistic for the p -dimensional case is

$$(1) \quad T_n = \sup_{a \neq 0} T(a'X_1, \dots, a'X_n)$$

We assume that the function T is location and scale invariant, i.e. that $T(d + cV_1, \dots, d + cV_n) = T(V_1, \dots, V_n)$ for all real $c > 0$ and real d and for all real $V_1, \dots, V_n$. Then, under H_0 , the distribution of T_n does not depend on μ and it depends on Σ only through its rank m , say. To see this, write $X_i = \mu + BZ_i$ with B $p \times m$ of rank m such that $BB' = \Sigma$ and $Z_1, \dots, Z_n$ i.i.d. $N_m(0, I)$ -distributed; then

$$\sup_{a \neq 0} T(a'X_1, \dots, a'X_n) = \sup_{\|b\|=1} T(b'Z_1, \dots, b'Z_n)$$

where the second sup is taken over all m -dimensional real vectors b such that $\|b\| = (b'b)^{1/2} = 1$. Consequently, to study the null distribution of (1) we can assume without loss of generality that the X_i 's are i.i.d. $N_p(0, I)$ -distributed and this assumption will be in effect for the rest of this paper unless otherwise stated.

Five special choices of T will be considered, viz.

- (a) the third moment measure of skewness,
- (b) three choices based on the fourth moment measure of kurtosis,
- (c) the statistic based on graphical methods proposed by de Wet and Venter (1972).

The first four of these statistics have essentially been suggested by Malkovich and Afifi (1973) who report on their power properties; we show how their asymptotic null distributions can be characterized and we estimate by Monte Carlo methods and graph these distributions for the cases $p = 2$ and 3 . The small sample null distributions are also estimated and compared with the asymptotic results.

The statistic proposed in de Wet and Venter (1972) is related to the Shapiro-Wilk test (see also Gregory (1977) and Durbin (1977)) which is known to have good power properties. Here we suggest a characterization of the asymptotic null distributions of the corresponding multivariate test which generalizes the corresponding univariate result but, unfortunately, is not suitable for computational purposes. The null distribution is estimated by Monte Carlo for the cases $p = 2, 3$ and graphed.

2. Tests based on skewness

In this paragraph we choose $T = T_1$ with

$$(2) \quad T_1(V_1, \dots, V_n) = n^{-1/2} \sum_{j=1}^n (V_j - \bar{V}_n)^3 / s^3$$

where

$$(3) \quad \bar{V}_n = n^{-1} \sum_{j=1}^n V_j, s^2 = n^{-1} \sum_{j=1}^n (V_j - \bar{V}_n)^2.$$

Letting

$$(4) \quad T_{1n}(a) = T_1(a'X_1, \dots, a'X_n)$$

the first test statistic for multivariate normality is

$$(5) \quad T_{1n} = \sup_{\|a\|=1} T_{1n}(a)$$

At first sight, a related statistic of interest is $\sup_{\|a\|=1} |T_{1n}(a)|$; however, since for any a we have $T_{1n}(-a) = -T_{1n}(a)$, we get

$$\sup_{\|a\|=1} T_{1n}(a) = \sup_{\|a\|=1} \max(T_{1n}(a), T_{1n}(-a)) = \sup_{\|a\|=1} \max(T_{1n}(a), -T_{1n}(a)) = \sup_{\|a\|=1} |T_{1n}(a)|,$$

so that in fact, this is not a new statistic.

To get the asymptotic distribution of T_{1n} we first remark that for $V_1, V_2, \dots$ i.i.d. $N(0,1)$ - distributed (also more generally, if only $EV_i = EV_i^3 = 0$, $EV_i^2 = 1$), we have

$$(6) \quad T_1(V_1, \dots, V_n) - (6/n)^{1/2} \sum_{j=1}^n h_3(V_j) \xrightarrow{P} 0$$

as $n \rightarrow \infty$, where

$$(7) \quad h_3(V) = (V^3 - 3V)/6^{1/2}$$

is the normalized third Hermitian polynomial. (For the sake of continuity with later results we work with the normalized Hermitian polynomials rather than the ordinary Hermitian polynomials which would have been somewhat simpler at this stage; the normalized m -th Hermitian polynomial will be denoted by h_m which is given by $h_m(x) = (2^m m!)^{-1/2} H_m(x/2^{1/2})$ with H_m the m -th Hermitian polynomial of Erdelyi (1953) Vol II.) We do know whether or not the asymptotic relation (6) is new; known facts such as the asymptotic normality of the third moment skewness and its asymptotic independence from the mean and variance follow immediately from it. If we define

$$(8) \quad T_{1n}^*(a) = (6/n)^{1/2} \sum_{j=1}^n h_3(a'X_j)$$

and

$$(9) \quad T_{1n}^* = \sup_{\|a\|=1} T_{1n}^*(a)$$

then (6) leads one to expect that

$$(10) \quad T_{1n} - T_{1n}^* \xrightarrow{P} 0;$$

we will show that this holds and will obtain the asymptotic distribution of T_{1n}^* ; it will

then follow that T_{1n} has the same asymptotic distribution.

For clarity of exposition we specialize now to the case $p = 2$; put $a' = (a_1 \ a_2)$ and $X'_j = (X_{1j} \ X_{2j})$; then from the identity

$$h_3(a_1 X_{1j} + a_2 X_{2j}) = a_1^3 h_3(X_{1j}) + 3^{1/2} a_1^2 a_2 h_2(X_{1j}) h_1(X_{2j}) + \\ 3^{1/2} a_1 a_2^2 h_1(X_{1j}) h_2(X_{2j}) + a_2^3 h_3(X_{2j})$$

for $a_1^2 + a_2^2 = 1$ (see Erdelyi (1953) Vol II p. 196) it follows that

$$(11) \quad T_{1n}^*(a) = b' W_n$$

where

$$(12) \quad b' = 6^{1/2} (a_1^3 \quad 3^{1/2} a_1^2 a_2 \quad 3^{1/2} a_1 a_2^2 \quad a_2^3)$$

$$(13) \quad W_n' = n^{-1/2} \sum_{j=1}^n (h_3(X_{1j}) \quad h_2(X_{1j}) h_1(X_{2j}) \quad h_1(X_{1j}) h_2(X_{2j}) \quad h_3(X_{2j}))$$

By the multivariate central limit theorem and the orthonormality properties of the Hermitian polynomials we find that

$$(14) \quad D(W_n) \rightarrow N_4(0, I)$$

where I is the 4×4 identity matrix.

Now, for w 4-dimensional real vectors, $\sup_{\|a\|=1} b'w$ is continuous in w ; hence

$$(15) \quad D(T_{1n}^*) \equiv D(\sup_{\|a\|=1} b'W_n) \rightarrow D(\sup_{\|a\|=1} b'W)$$

where W is $N_4(0, I)$ distributed. We have

$$(16) \quad T_{1n}(a) = T_{1n}^*(a) + R_n(a)$$

where

$$(17) \quad R_n(a) = T_{1n}^*(a) [(a' S_n a)^{-3/2} - 1] + \\ (a' S_n a)^{-3/2} [2n^{1/2} (a' \bar{X}_n)^3 - 3a' \bar{X}_n n^{-1/2} \sum_{j=1}^n [(a' X_j)^2 - 1]]$$

where

$$S_n = n^{-1} \sum_{j=1}^n (X_j - \bar{X}_n)(X_j - \bar{X}_n)', \quad \bar{X}_n = n^{-1} \sum_{j=1}^n X_j.$$

Using the law of large numbers and (15) together with a similar result for

$$\sup_{\|a\|=1} n^{-1/2} \sum_{j=1}^n [(a'X_j)^2 - 1],$$

it follows that

$$\sup_{\|a\|=1} |R_n(a)| \xrightarrow{P} 0.$$

Thus (10) follows and we have proved that

$$(18) \quad D(T_{1n}) \rightarrow D(\sup_{\|a\|=1} b'W)$$

where b is a vector function of a given by (12).

Extension of the argument to the cases $p > 2$ is straightforward: (18) holds with b and W k -dimensional vectors where

$$(19) \quad k = \frac{1}{6} p(p+1)(p+2)$$

and

$$(20) \quad b' = 6^{1/2} (b^{(1)'} \quad b^{(2)'} \quad b^{(3)'})$$

with

$$(21) \quad b^{(1)'} = (a_1^3 \quad a_2^3 \quad \dots \quad a_p^3) \quad (p\text{-dimensional})$$

$$b^{(2)'} = 3^{1/2} (a_1^2 a_2 \quad a_1 a_2^2 \quad \dots \quad a_{p-1} a_p^2) \quad (p(p-1)\text{-dimensional})$$

$$b^{(3)'} = 6^{1/2} (a_1 a_2 a_3 \quad \dots \quad a_{p-2} a_{p-1} a_p) \quad (\binom{p}{3}\text{-dimensional})$$

$$(22) \quad W \text{ is } N_k(0, I)\text{-distributed}$$

Unfortunately it does not seem possible to compute $D(\sup_{\|a\|=1} b'W)$ exactly; however, it can be estimated by Monte Carlo methods.

For $p = 2$ the procedure is as follows:

Generate M realizations of the random vector W ; take $a' = (a_1 \ a_2) = (\cos\theta \ \sin\theta)$; it saves computer time to estimate the distribution of $\sup_{0 \leq \theta < \pi} |b(\theta)'W|$ which is the same

as that of $\sup_{0 \leq \theta < 2\pi} b(\theta)'W$; here $b(\theta)$ is the vector given by (12) with a as above. For K equally spaced values θ_j of θ in $[0, \pi)$ compute the corresponding values $b_j = b(\theta_j)$ of b from (12) and then for each realization of W find $\max_j |b_j'W|$; for K large enough these serve as close approximations to realizations of the random variable $\sup_{\|a\|=1} b'W$ and with M large the empirical distribution function (e.d.f.) of these approximate realizations serves as estimate of $D(\sup_{\|a\|=1} b'W)$. Experimentation with an increasing sequence of values of K showed that with $M = 20\,000$ repetitions the resulting estimate based on $K > 50$ differs insignificantly from that for $K = 50$. The e.d.f. was computed at approximately 30 points and then smoothed by third degree interpolation; it was plotted by computer driven plotter and further smoothed by hand by a graphic artist. The curve marked ∞ in Figure 1.2 shows the final result. We also estimated the exact distribution of T_{1n} for $n = 10, 20, 50$ and 100 in the same manner (for $n \leq 50$ M was $20\,000$ and for $n = 100$ M was only $2\,000$) and these curves are also shown in Figure 1.2. In order to display more detail in the upper percentile region the vertical scale is that of normal probability paper. It is clear that the asymptotic distribution provides a fair approximation to the exact distribution for sample sizes $n > 50$.

The procedure for $p = 3$ is similar but somewhat more involved: we take $a' = (a_1 \ a_2 \ a_3) = (\cos\phi\sin\theta \ \sin\phi\sin\theta \ \cos\theta)$ with $0 \leq \theta \leq \pi/2$ and $0 \leq \phi < 2\pi$ and vary (θ, ϕ) over a grid of points such that a varies over a grid of K approximately equally spaced points on the upper half of the unit sphere in three dimensions. Figure 1.3 shows the estimated asymptotic distribution together with estimates of the exact distribution for sample sizes $n = 10, 20, 50$ and 100 ; these are based on $K = 610$ points and $M = 20\,000$ repetitions for $n = 10$; $M = 15\,000$ repetitions for $n = 20$; $M = 5\,000$ repetitions for $n = 50$; $M = 2\,000$ for $n = 100$ and $M = 20\,000$ repetitions for the asymptotic distribution. Again the asymptotic distribution gives a useful approximation for $n > 50$.

Remarks

1. $T_{1n}^*(a)$ may be viewed as a stochastic process with parameter a varying over the set $\|a\| = 1$; from (8) the joint distribution of $T_{1n}^*(a)$ and $T_{1n}^*(c)$ converges to the bivariate normal with zero mean and covariance matrix

$$\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

where

$$\rho = \rho(a, c) = E h_3(a'X_1)h_3(c'X_1) = (a'c)^3$$

Extending we find that the finite dimensional distributions of the process converge to those of a zero mean Gaussian process $Y_3(a)$, say, with covariance function

$6\rho(a,c)$ as above; this makes plausible the suggestion that the process $T_{1n}(a)$ converges weakly to $Y_3(a)$ and that $D(T_{1n}) = D(\sup_{\|a\|=1} T_{1n}(a)) \rightarrow D(\sup_{\|a\|=1} Y_3(a))$. Now, in fact, $Y_3(a) = b'W$ with b given by (21) and W distributed as in (22) above is a Gaussian process with covariance function ρ so that our exposition amounts to a direct verification of this suggestion.

2. A bound on $D(\sup_{\|a\|=1} b'W)$ can be obtained as follows:

By the Schwarz inequality we have $|b'W| \leq (b'b)^{1/2} (W'W)^{1/2}$:

But $b'b = 6(a_1^2 + \dots + a_p^2)^3 = 6$ and $W'W$ is χ_k^2 -distributed; hence

$$P(\sup_{\|a\|=1} b'W \leq x) \geq P(\chi_k^2 \leq x^2/6)$$

so that the α -th percentile of $D(\sup_{\|a\|=1} b'W)$, ξ_α , say, is bounded by

$$(23) \quad \xi_\alpha \leq (6\chi_k^2(\alpha))^{1/2}$$

with $\chi_k^2(\alpha)$ the α -th percentile of the χ_k^2 -distribution. For comparison purposes a few values of the two sides of (23) are shown in the following table, ξ_α having been read from Figures 1.2 and 1.3:

p	α	ξ_α	$(6\chi_k^2(\alpha))^{1/2}$
2	,95	6,5	7,54
	,99	7,8	8,93
3	,95	7,8	10,48
	,99	9,0	11,80

3. Tests based on kurtosis

Here we look at

$$(24) \quad T_2(V_1, \dots, V_n) = n^{-1/2} \sum_{j=1}^n [(V_j - \bar{V}_n)^4 T_s^4 - 3]$$

and with

$$(25) \quad T_{2n}(a) = T_2(a'X_1, \dots, a'X_n)$$

we are interested in the test statistics

$$(26) \quad T_{2n} = \sup_{\|a\|=1} T_{2n}(a)$$

$$(27) \quad T_{3n} = \sup_{\|a\|=1} (-T_{2n}(a))$$

$$(28) \quad T_{4n} = \sup_{\|a\|=1} |T_{2n}(a)|$$

A companion to (6) is the asymptotic relation

$$(20) \quad T_2(V_1, \dots, V_n) - (24/n)^{1/2} \sum_{j=1}^n h_4(V_j) \xrightarrow{P} 0$$

with

$$(30) \quad h_4(V) = (V^4 - 3V^2 + 6)/24^{1/2}$$

Hence the indication is that e.g. T_{2n} will have the same asymptotic distribution as

$$\sup_{\|a\|=1} (24/n)^{1/2} \sum_{j=1}^n h_4(a'X_j)$$

and this can be shown by arguments similar to those used in § 2. The results are

$$(31) \quad D(T_{2n}) \rightarrow D\left(\sup_{\|a\|=1} b'W\right)$$

$$D(T_{3n}) \rightarrow D\left(\sup_{\|a\|=1} (-b'W)\right)$$

$$D(T_{4n}) \rightarrow D\left(\sup_{\|a\|=1} |b'W|\right)$$

where the dimension of the vectors b and W is

$$(34) \quad k = \frac{1}{24} p(p+1)(p+2)(p+3)$$

and

$$b' = 24^{1/2} (b^{(1)'} \quad b^{(2)'} \quad b^{(3)'} \quad b^{(4)'} \quad b^{(5)'})$$

with

$$b^{(1)'} = (a_1^4 \dots a_p^4) \quad (p\text{-dimensional})$$

$$b^{(2)'} = 2(a_1^3 a_2 \dots a_{p-1} a_p^3) \quad (2\binom{p}{2}\text{-dimensional})$$

$$b^{(3)'} = 6^{1/2} (a_1^2 a_2^2 \dots a_{p-1}^2 a_p^2) \quad \left(\binom{p}{2}\text{-dimensional}\right)$$

$$b^{(4)'} = 12^{1/2} (a_1^2 a_2^2 a_3^2 \dots a_{p-2}^2 a_{p-1}^2 a_p^2) \quad \left(3\binom{p}{3}\text{-dimensional}\right)$$

$$b^{(5)'} = 24^{1/2} (a_1 a_2 a_3 a_4 \dots a_{p-3} a_{p-2} a_{p-1} a_p) \quad \left(\binom{p}{4}\text{-dimensional}\right)$$

and

W is $N_k(0, I)$ -distributed.

(Of course, for $p = 2$ the last two partitions and for $p = 3$ the last partition will not be present.)

These asymptotic distributions can be estimated as outlined before; Figures 2.2, 3.2, 4.2 and 2.3, 3.3, 4.3 show the results for the cases $p = 2$ and 3 respectively. In the cases $n = 10, 20, 50$ and ∞ , $M = 20\,000$ repetitions were used but for $n = 100$ M was taken 2 000 only. It is clear that the asymptotic distributions approximate the exact distributions much poorer in these cases, especially in the high percentile regions; this finding appears to be consistent with the statement of D'Agostino and Tietjen (1971) (p. 670) for the case $p = 1$, viz. "... the extremely long thin tail, i.e. the upper tail, of b_2 for $n < 50$."

Remarks analogous to those at the end of § 3 are also appropriate here.

4. Test based on graphical methods

The choice of T proposed in de Wet and Venter (1972) is

$$(36) \quad T(V_1, \dots, V_n) = \sum_{j=1}^n [(V_{jn} - \bar{V}_n)/s - H_{jn}]^2 - a_n$$

where $V_{1n}, \dots, V_{nn}$ are the order statistics of $V_1, \dots, V_n$ and $H_{jn} = \Phi^{-1}(j/(n+1))$ with Φ the $N(0, 1)$ -distribution function and where for our present purposes we take

$$(37) \quad a_n = (n+1)^{-1} \sum_{k=1}^n c_{kn} - 1,5$$

with

$$(38) \quad c_{kn} = \phi(H_{kn})^{-2} (k(n+1))(1 - k/(n+1))$$

and ϕ the $N(0, 1)$ density. Table 1 lists a_n for some values of n .

Let

$$(39) \quad T_n(a) = T(a'X_1, \dots, a'X_n).$$

Then the test for multivariate normality corresponding to the univariate test (36) is

$$(40) \quad T_{5n} = \sup_{\|a\|=1} T_n(a).$$

To get a characterization of its asymptotic distribution analogous to those above, we can proceed as follows.

It is shown in de Wet and Venter (1972) that $D(T(V_1, \dots, V_n)) \rightarrow D(\sum_{m=3}^{\infty} (Y_m^2 - 1)/m)$ with $Y_3, Y_4, \dots$ i.i.d. $N(0,1)$; for our present purposes we need the stronger results of Gregory (1977). As in the proof of his Theorem 3.1 it can be shown that

$$(41) \quad T(V_1, \dots, V_n) - n^{-1} \sum_{i \neq j} Q(U_i, U_j) \xrightarrow{P} 0$$

where $U_i = \Phi(V_i)$ so that $U_1, \dots, U_n$ are i.i.d. uniformly distributed on $(0,1)$. Also Q is given by

$$(42) \quad Q(u,v) = Q_0(u,v) - h_1(\Phi^{-1}(u))h_1(\Phi^{-1}(v)) - \frac{1}{2}h_2(\Phi^{-1}(u))h_2(\Phi^{-1}(v))$$

where

$$(43) \quad Q_0(u,v) = \int_0^1 [I(u \leq t) - t][I(v \leq t) - t] \phi(\Phi^{-1}(t))^{-2} dt$$

with $I(S)$ the indicator of the set S and $h_1(x) = x$, $h_2(x) = (x^2 - 1)/2^{1/2}$. Now let a_1 and a_2 be two p -dimensional vectors with $\|a_1\| = 1 = \|a_2\|$; with $U_{1i} = \Phi(a_1' X_i)$ and $U_{2i} = \Phi(a_2' X_i)$ it then follows from (41) that

$$(44) \quad \begin{pmatrix} T_n(a_1) \\ T_n(a_2) \end{pmatrix} - \begin{pmatrix} n^{-1} \sum_{i \neq j} Q(U_{1i}, U_{1j}) \\ n^{-1} \sum_{i \neq j} Q(U_{2i}, U_{2j}) \end{pmatrix} \xrightarrow{P} 0$$

Extending somewhat the argument of the proof of Theorem 2.1 of Gregory (1977a) we can show that

$$(45) \quad D\left(\begin{pmatrix} n^{-1} \sum_{i \neq j} Q(U_{1i}, U_{1j}) \\ n^{-1} \sum_{i \neq j} Q(U_{2i}, U_{2j}) \end{pmatrix}\right) \rightarrow D\left(\begin{pmatrix} \sum_{m=3}^{\infty} (Y_{1m}^2 - 1)/m \\ \sum_{m=3}^{\infty} (Y_{2m}^2 - 1)/m \end{pmatrix}\right)$$

with

$$\begin{pmatrix} Y_{1m} \\ Y_{2m} \end{pmatrix} \text{ i.i.d. } N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_m \\ \rho_m & 1 \end{pmatrix} \right)$$

where

$$(46) \quad \rho_m = E h_m(a'_1 X_1) h_m(a'_2 X_1).$$

We outline this argument briefly: It is shown in de Wet and Venter (1972) that the eigenvalues of Q are $\gamma_m = 1/m$, $m = 3, 4, \dots$ with corresponding eigenfunctions $h_m(\Phi^{-1}(u))$, $m = 3, 4, \dots$ Hence we have the bilinear expansion $Q(u, v) \approx \sum \gamma_m h_m(\Phi^{-1}(u)) h_m(\Phi^{-1}(v))$ so that

$$n^{-1} \sum_{i \neq j} Q(U_{ri}, U_{rj}) \approx \sum_m \gamma_m (\xi_{rnm}^2 - c_{rnm})$$

with

$$\xi_{rnm} = n^{-1/2} \sum_{i=1}^n h_m(a'_r X_i), c_{rnm} = n^{-1} \sum_{i=1}^n h_m(a'_r X_i)^2$$

for $r = 1, 2$. By the multivariate central limit theorem it follows that for $k \neq m$

$$D \left(\begin{pmatrix} \xi_{1nm} \\ \xi_{2nm} \\ \xi_{1nk} \\ \xi_{2nk} \end{pmatrix} \right) \rightarrow N_4 \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_m & 0 & 0 \\ \rho_m & 1 & 0 & 0 \\ 0 & 0 & 1 & \rho_k \\ 0 & 0 & \rho_k & 1 \end{pmatrix} \right)$$

and by the law of large numbers $c_{rnm} \rightarrow 1$ as $n \rightarrow \infty$ a.s. indicating the validity of (45).

Returning to (46), since $a'_1 X_1$ and $a'_2 X_1$ are jointly $N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & a'_1 a'_2 \\ a'_1 a'_2 & 1 \end{pmatrix} \right)$

distributed use of the Mehler identity and the orthonormality properties of the Hermitian polynomials lead to

$$(47) \quad \rho_m = (a'_1 a'_2)^m.$$

Introduce a sequence of independent Gaussian processes $\{Y_m(a)\}$ with parameter a varying over the set $\|a\| = 1$ in p dimensions, having zero means and covariance functions

$$(48) \quad E Y_m(a_1) Y_m(a_2) = (a'_1 a'_2)^m;$$

then the above arguments suggest that the sequence of stochastic processes $T_n(a)$ converges weakly to the process

$$(49) \quad Y(a) = \sum_{m=3}^{\infty} (Y_m(a)^2 - 1)/m$$

and that

$$(50) \quad D(T_{5n}) = D(\sup_{\|a\|=1} T_n(a)) \rightarrow D(\sup_{\|a\|=1} Y(a)).$$

This result is interesting in that it nicely generalizes to the multivariate case the type of characterization of asymptotic distributions one usually gets in the univariate case. However, it does not seem amenable to actual computing of the asymptotic distribution.

A possible approach to estimation of the asymptotic distribution is suggested by our earlier results. With

$$(51) \quad Y_{nm}(a) = n^{-1/2} \sum_{i=1}^n h_m(a'X_i)$$

the process $Y_m(a)$ is the weak limit as $n \rightarrow \infty$ of the process $Y_{nm}(a)$; now, for $\|a\| = 1$, by the addition formula for Hermitian polynomials (Erdélyi (1953) Vol. II p. 196),

$$(52) \quad h_m(a'X_i) = (m!)^{1/2} r_1 + \dots + r_p = m \frac{a_1^{r_1} \dots a_p^{r_p}}{(r_1! \dots r_p!)^{1/2}} h_{r_1}(X_{1i}) \dots h_{r_p}(X_{pi})$$

Hence

$$(53) \quad Y_{nm}(a) = b_m' W_{nm}$$

where b_m is a $k_m = \binom{m+p-1}{m}$ -dimensional vector depending on a having typical component

$a_1^{r_1} \dots a_p^{r_p} (m!/r_1! \dots r_p!)^{1/2}$ and W_{nm} is a random vector of the same dimension having corresponding typical component

$$(54) \quad n^{-1/2} \sum_{i=1}^n h_{r_1}(X_{1i}) \dots h_{r_p}(X_{pi});$$

then $D(W_{nm}) \rightarrow D(W_m) = N_{k_m}(0, I)$; hence the process $Y_m(a)$ can be represented as

$Y_m(a) = b_m' W_m$. (Alternatively, check directly that this process has the desired properties). These processes are the higher order analogues of $Y_3(a)$ and $Y_4(a)$ already encountered above. In principle one could now generate realizations of the vectors W_m and hence of the processes $Y_m(a)$ and $Y(a)$; then maximization by numerical search would lead to realizations of $\sup_{\|a\|=1} Y(a)$ enabling estimation of its distribution. Two difficulties militate against this program: the dimensions of the vectors involved grow fast with increasing m and p and the series (49) converges slowly. The cases $p = 2$ and 3

might not be hopeless but we have not attempted it; in stead we directly estimated by simulation the distributions of T_{5n} for $n = 10, 20, 50$ and 100 . This was done as described above; for $n = 10, 20, 50$ $M = 20\,000$ repetitions were used but for $n = 100$ only $M = 2\,000$ repetitions were used. Figures 5.2 and 5.3 display the results. It is evident that for practical purposes convergence to the limiting distribution has occurred at $n = 50$.

TABLE 1
The constants a_n

n	a_n
5	0,06
10	0,36
20	0,62
30	0,75
40	0,84
50	0,90
100	1,08
200	1,23
400	1,37
600	1,44
800	1,49
1 000	1,53

FIGURE 1.2
Distribution function of T_{1n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

Fig. ..2

Distribution function of T_{1n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

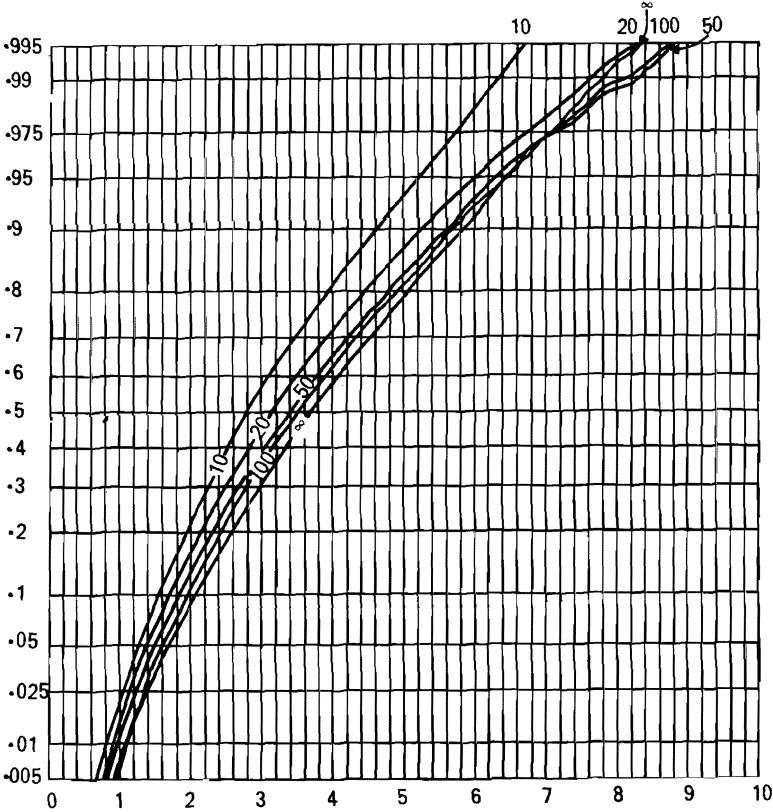


FIGURE 1.3
Distribution function of T_{1n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

Fig. 1.3

Distribution function of T_{1n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

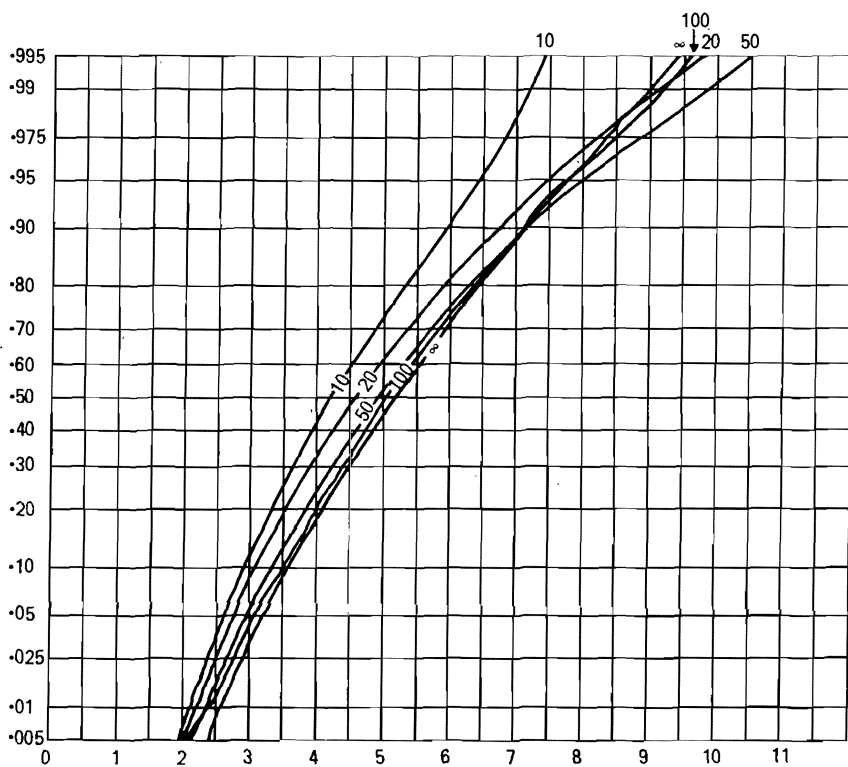


FIGURE 2.2
Distribution function of T_{2n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

Fig.2.2

Distribution function of T_{2n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

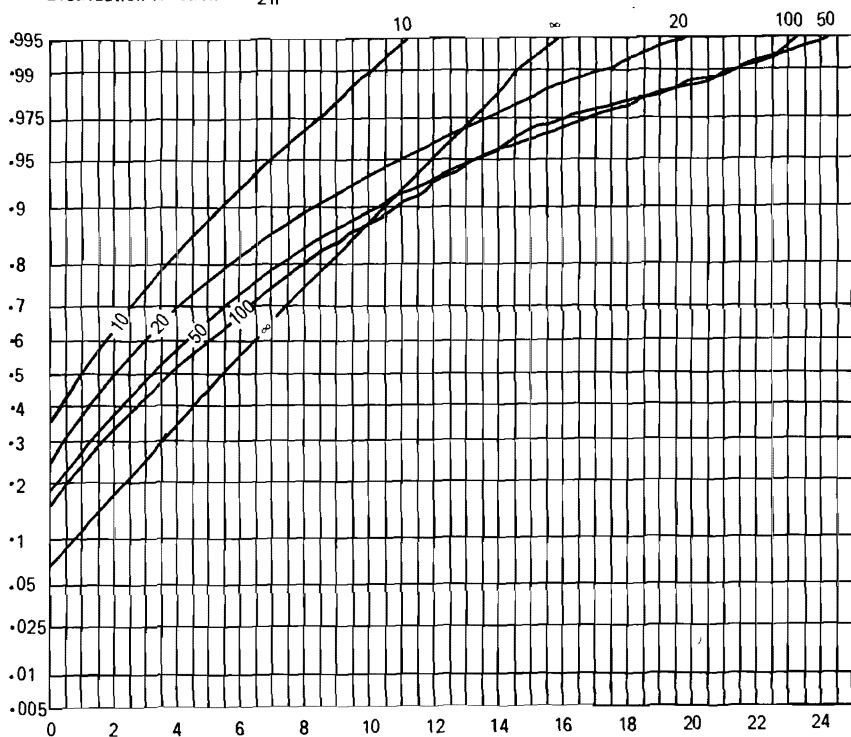


FIGURE 2.3
Distribution function of T_{2n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

Fig.2.3

Distribution function of T_{2n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

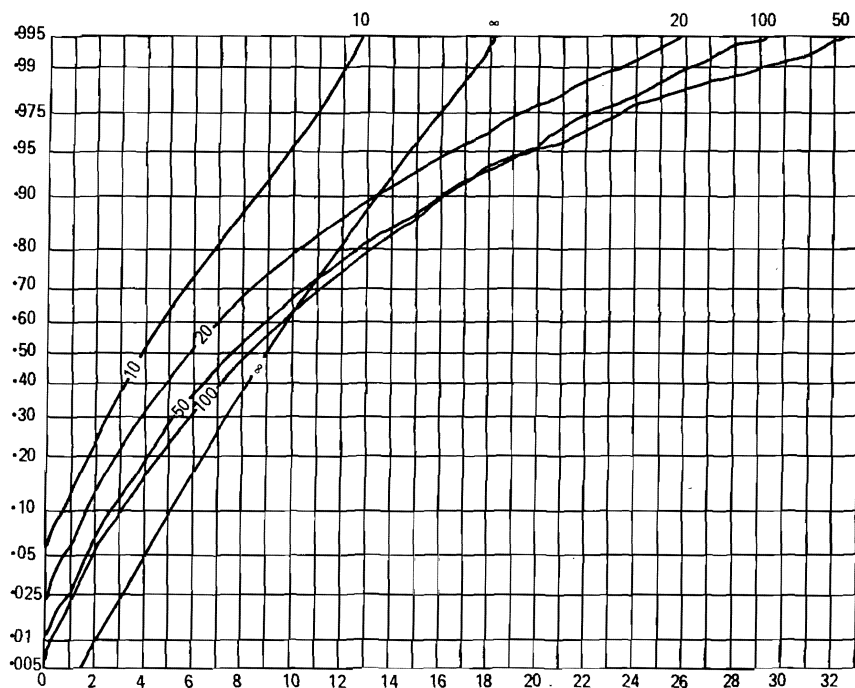


FIGURE 3.2
 Distribution function of T_{3n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

Fig.3.2

Distribution function of T_{3n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

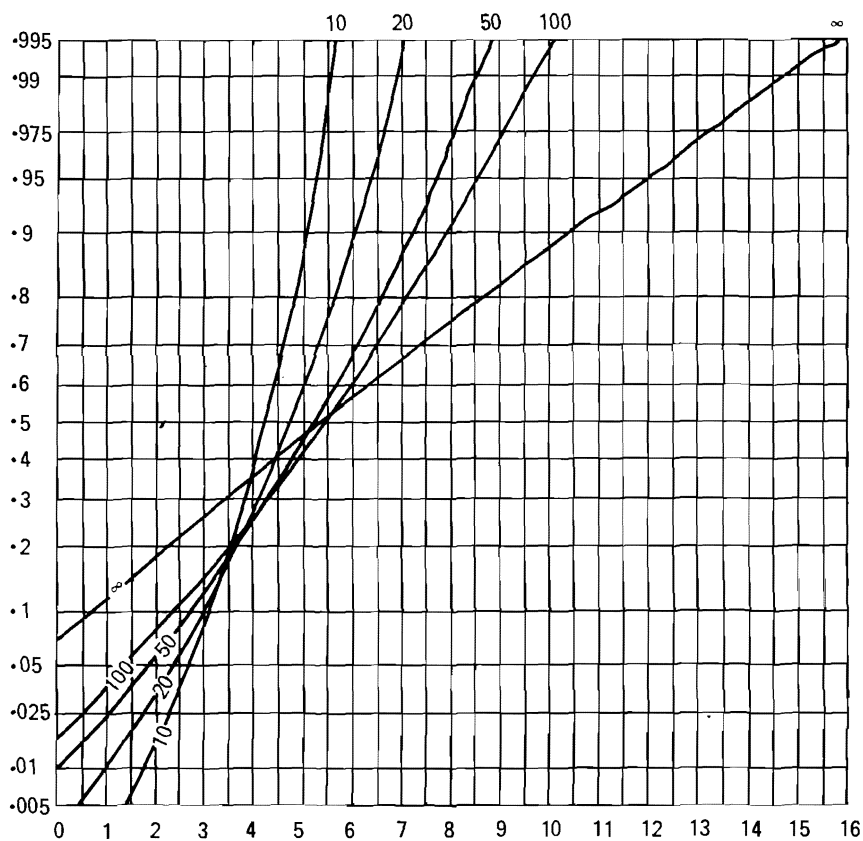


FIGURE 3.3
Distribution function of T_{3n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

Fig. 3.3

Distribution function of T_{3n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

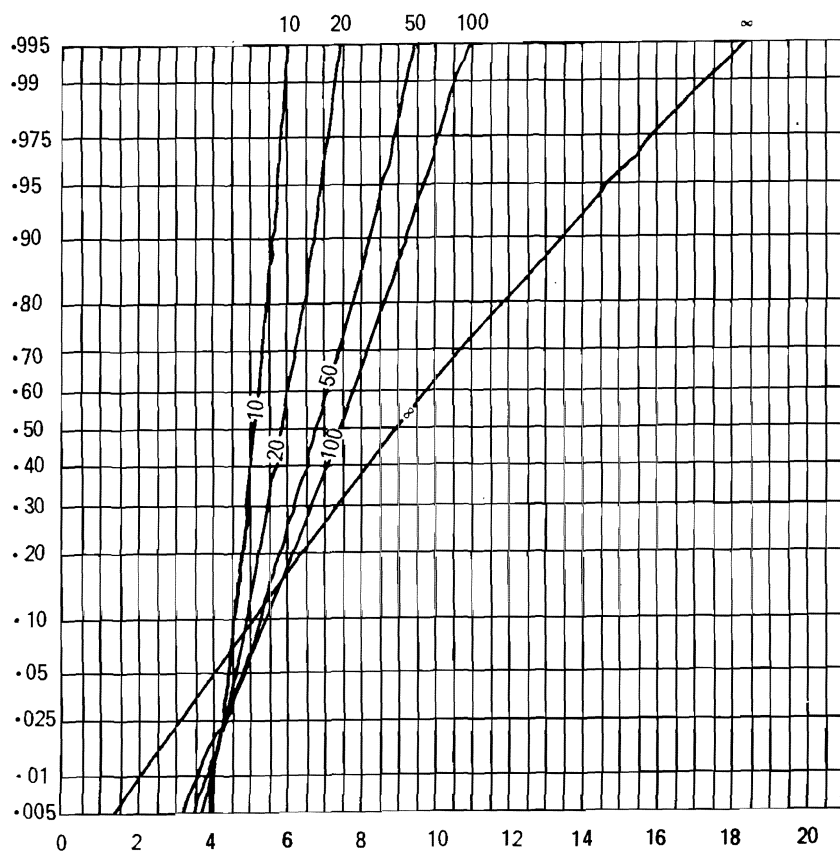


FIGURE 4.2
Distribution function of T_{4n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

Fig.4.2

Distribution function of T_{4n} for $n = 10, 20, 50, 100, \infty$ and $p = 2$

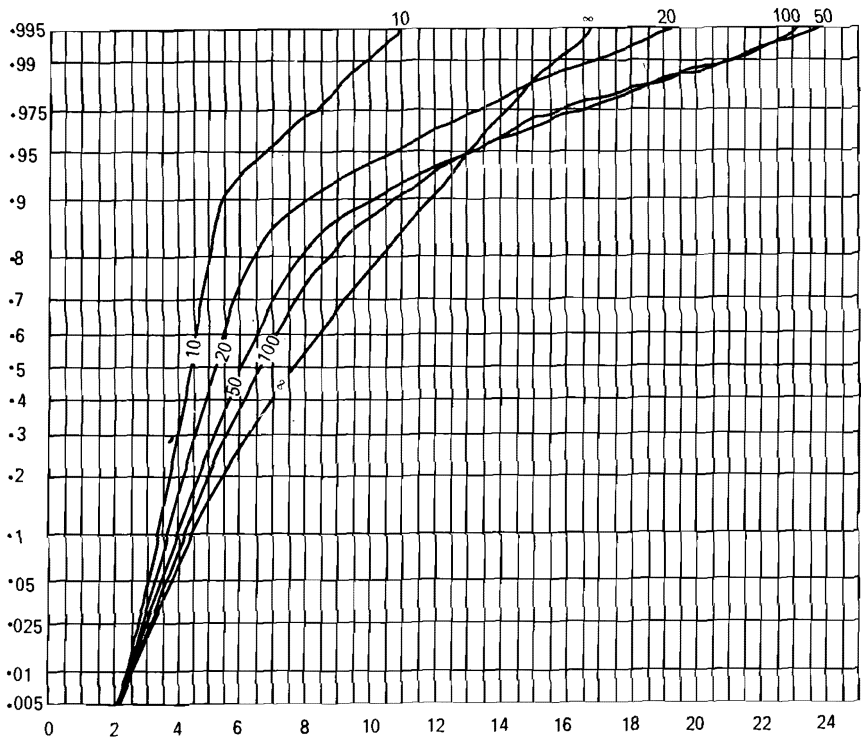


FIGURE 4.3
 Distribution function of T_{4n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

Fig.4.3

Distribution function of T_{4n} for $n = 10, 20, 50, 100, \infty$ and $p = 3$

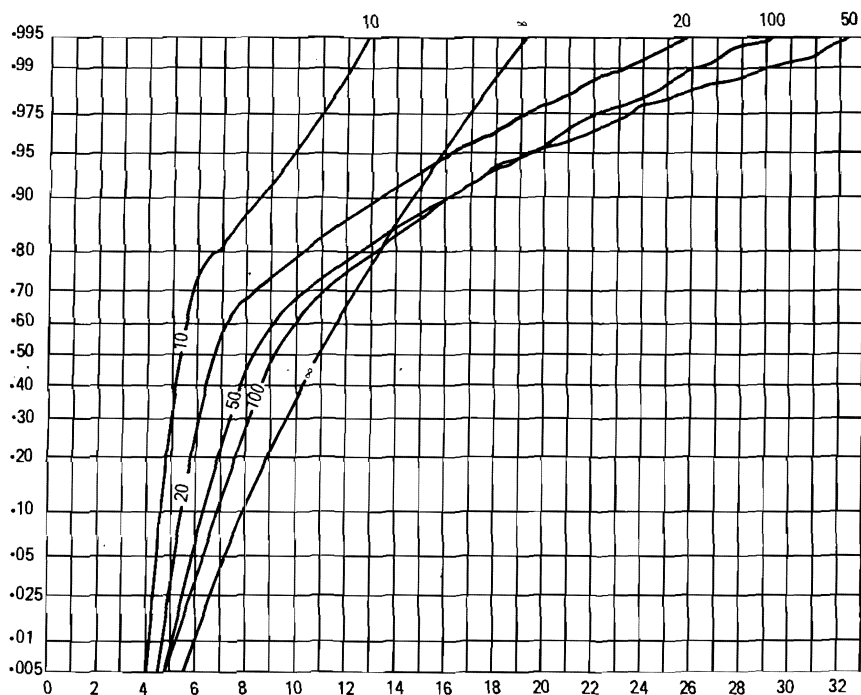


FIGURE 5.2
Distribution function of T_{5n} for $n = 10, 20, 50, 100$, and $p = 2$

Fig. 5.2

Distribution function of T_{5n} for $n = 10, 20, 50, 100$ and $p = 2$

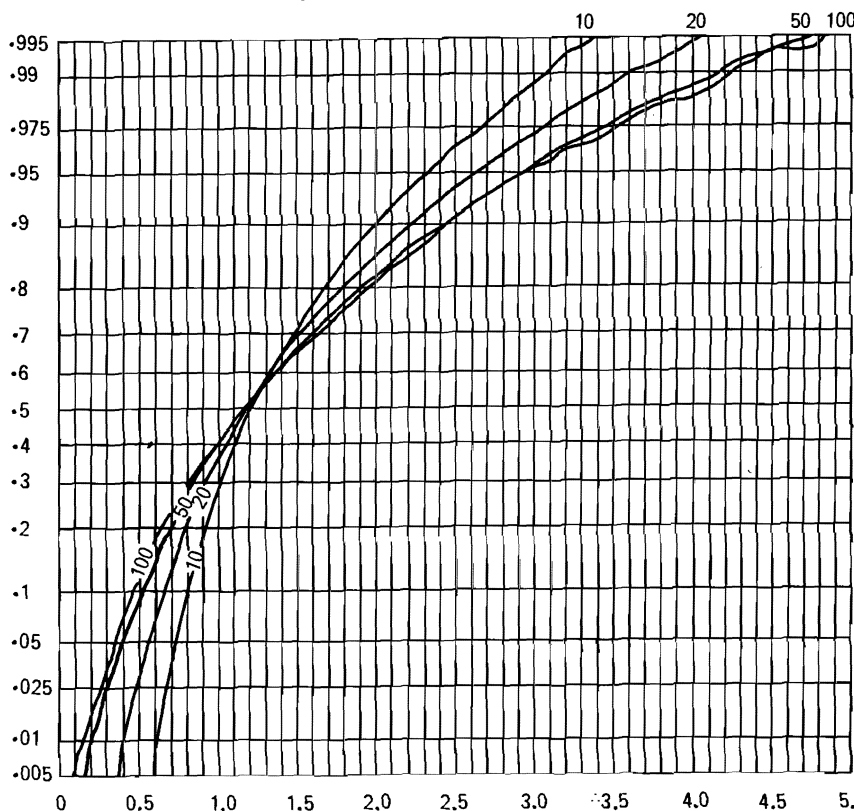
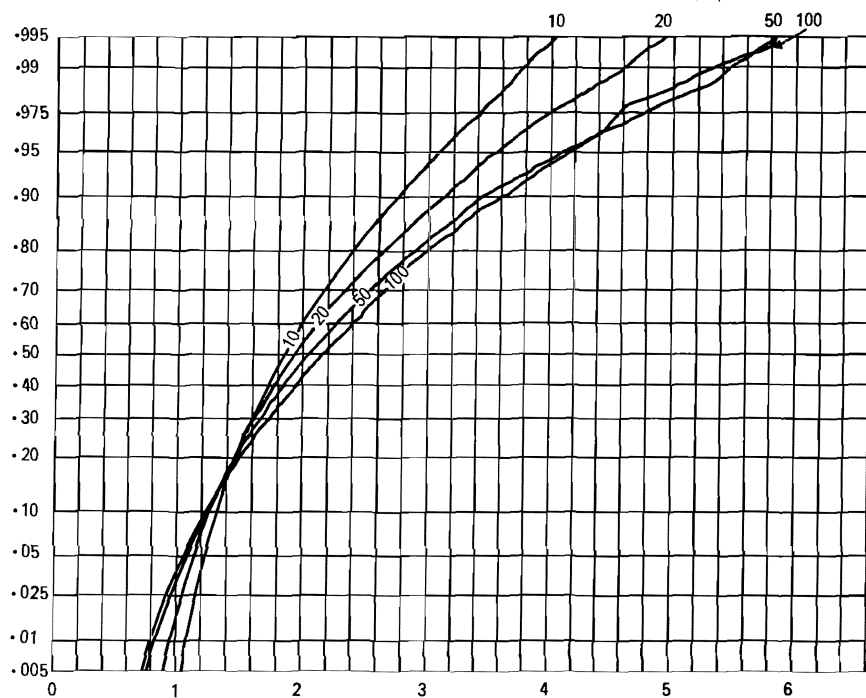


FIGURE 5.3
Distribution function of T_{5n} for $n = 10, 20, 50, 100$ and $p = 3$

Fig.5.3

Distribution function of T_{5n} for $n = 10, 20, 50, 100$ and $p = 3$



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